HaloFormCraft
Technical Documentation

Current Application & System Specification

Generated for Internal Audit - May 2026

Multi-Center Data Management Strategy

1. Overview

As HaloFormCraft moves into production and field testing across multiple hospital sites, a structured hierarchy is required to ensure data integrity, facilitate multi-site management, and prevent data "contamination" (mixing data from different centers).

This strategy outlines the transition to a **Multi-Tenant, Project-Centric Hierarchy**. A single project may involve multiple hospitals, centers, camps, or field locations participating in the same data collection workflow.

---

2. The Organizational Hierarchy

To scale from 3 centers to 30+, the application will follow a three-tier model where the **Project** is the central anchor:

Tier 1: Project (The Root)

- **Definition:** A specific clinical study or nationwide/statewide operational workflow.
- **Example:** "Longevity Pilot 2024", "Maternal Health Screening".
- **Logic:** A single Project can have multiple Hospitals mapped to it.

Tier 2: Hospital/Center (The Participant)

- **Definition:** The physical facility, institution, or field location participating in the study.
- **Example:** Baptist Hospital, M.S. Ramaiah, SMISMSR.
- **ID Strategy:** Each hospital is assigned a **Short Code** (e.g., `BPA`, `MSR`).
- **Logic:** Hospitals use assigned forms within their project scope.

Tier 3: Form (The Tool)

- **Definition:** The actual digital data collection instrument.
- **Logic:** Forms are created and attached to the Project (or specifically to a Hospital context). Reusable across hospitals within the project.

---

3. Technical Implementation: The "Sorting Hat" Logic

To manage this in a single database, we use **Metadata Tagging**.

A. Site Codes vs. Names

- **Strategy:** Always use unique, short, alphanumeric codes (Slugs) for database keys.
- **Rationale:** Hospital names are for display (UI); Site Codes are for logic (DB). Names can have typos or change; codes are immutable.

B. Submission Flow (Data Tagging)

When a data collector opens a form at a center, the system automatically captures the origin: 1. **URL Parameter:** `https://haloform.com/fill/form_id?project=LON_2024&site=BPA&center=camp_A` 2. **Metadata Injection:** The UI reads the parameters and silently attaches them to the submission JSON.

C. Patient UI Contextualization (Dashboard Filtering)

To ensure patients only see relevant forms, the Patient Dashboard becomes "Context-Aware": 1. **Dynamic Filtering:** If a patient visits `patient.haloform.com?project=LON_2024&site=BPA`, the dashboard will only fetch and display forms belonging to that project and hospital. 2. **Context Locking:** The app saves the context in `localStorage`. If the patient refreshes the page or returns later, the app remains "locked" to that portal. 3. **Site-Specific Branding:** The UI can dynamically load the project's or hospital's logo, contact information, and welcome message based on the active context.

D. Database Storage Structure

Final submissions will be stored with a unified metadata block for better segregation and tracking:

```
{
"metadata": {
"project_id": "proj_001",
"hospital_id": "BPA",
"center_id": "camp_A",
"submitted_by": "user_123",
"submitted_at": "2026-05-13T10:00:00Z"
},
"data": {
"patient_age": 22,
"bmi": 19.4
}
}
```

E. QR Generation & Deployment Workflow

The Admin Panel serves as the central hub for generating physical access points for the field sites: 1. **Portal QRs:** A single QR code for a hospital under a project (e.g., `?project=LON_2024&site=BPA`) that directs patients to a filtered dashboard. 2. **Form-Specific QRs:** Deep-link QRs (e.g., `/fill/form_id?project=LON_2024&site=BPA`) that take the patient directly to a specific instrument while still preserving the site-tagging metadata. 3. **Automated Poster Generation:** The Admin UI will provide a "Print Site Poster" feature, generating a standardized PDF with the branding, instructions, and the encoded QR code for immediate physical deployment.

---

4. Key Benefits

Scalability & Management

Better scalability for multi-hospital deployments and easier management of nationwide/statewide programs. A new hospital can be simplified onboarded into an existing project.

Centralized Reporting & Data Segregation

Centralized project-level reporting and analytics. Better segregation and tracking of data by Project, Hospital, Center/Camp, and User.

Reusable Forms

Forms can be created once at the Project level and reused across all participating hospitals.

Data Isolation & Security

By tagging every row with `project_id` and `hospital_id`, we can easily implement **Row-Level Security**. A researcher from Ramaiah will only be able to query rows where `hospital_id == 'MSR'`.

---

5. Future Roadmap: PostgreSQL Integration

While SQLite works for local testing, the **PostgreSQL** migration will formalize these relationships using Foreign Keys: * `hospitals.project_id` $\rightarrow$ `projects.id` * `forms.project_id` $\rightarrow$ `projects.id` * `submissions.hospital_id` $\rightarrow$ `hospitals.id` * `submissions.project_id` $\rightarrow$ `projects.id`

This ensures that data cannot be orphaned and maintains the "Traceability Chain" required for clinical validation.

---

6. End-to-End Workflow: From Admin to Database

Step 1: Project Creation

BPA Admin creates a **Project** (e.g., a new nationwide program). * **Input:** "Longevity Pilot 2026" * **DB Result:** Row created in `projects` table.

Step 2: Hospital Mapping

Hospitals/Centers are mapped to the Project. * **Input:** Add "Baptist Hospital Bangalore" (Code: `BPA`) to the Project. * **DB Result:** Row created in `hospitals` table, linked to the `project_id`.

Step 3: Form Assignment

Forms are created and attached to the Project (or Hospital context). Hospitals use assigned forms within their project scope. * **DB Result:** `forms` table row is created with `project_id`.

Step 4: QR Generation (Automatic)

Upon clicking **"Publish"** in the Form Builder: 1. The system constructs a **Context-Aware URL**: `https://patient.haloform.com/fill/5?project=LON_2026&site=BPA` 2. The Admin UI renders this as a downloadable **QR Code** and a printable **Site Poster**.

Step 5: Patient Submission

1. Patient scans the QR code.
2. The Patient App extracts `project=LON_2026` and `site=BPA` from the URL.
3. Upon submission, the app attaches these IDs, along with `center_id` and `submitted_by`, to the metadata.
4. **Backend Result:** A new row in `submissions` is created with all clinical data perfectly tagged for sorting by project, hospital, center, and user.

---

7. System Flow Diagram

Multi-Center Data Management Strategy

1. Overview
As HaloFormCraft moves into production and field testing across multiple hospital sites, a structured hierarchy is required to ensure data integrity, facilitate multi-site management, and prevent data "contamination" (mixing data from different centers).

This strategy outlines the transition to a **Multi-Tenant, Project-Centric Hierarchy**. A single project may involve multiple hospitals, centers, camps, or field locations participating in the same data collection workflow.

---

2. The Organizational Hierarchy
To scale from 3 centers to 30+, the application will follow a three-tier model where the **Project** is the central anchor:

Tier 1: Project (The Root)
* **Definition:** A specific clinical study or nationwide/statewide operational workflow.
* **Example:** "Longevity Pilot 2024", "Maternal Health Screening".
* **Logic:** A single Project can have multiple Hospitals mapped to it.

Tier 2: Hospital/Center (The Participant)
* **Definition:** The physical facility, institution, or field location participating in the study.
* **Example:** Baptist Hospital, M.S. Ramaiah, SMISMSR.
* **ID Strategy:** Each hospital is assigned a **Short Code** (e.g., `BPA`, `MSR`).
* **Logic:** Hospitals use assigned forms within their project scope.

Tier 3: Form (The Tool)
* **Definition:** The actual digital data collection instrument.
* **Logic:** Forms are created and attached to the Project (or specifically to a Hospital context). Reusable across hospitals within the project.

---

3. Technical Implementation: The "Sorting Hat" Logic
To manage this in a single database, we use **Metadata Tagging**.

A. Site Codes vs. Names
* **Strategy:** Always use unique, short, alphanumeric codes (Slugs) for database keys.
* **Rationale:** Hospital names are for display (UI); Site Codes are for logic (DB). Names can have typos or change; codes are immutable.

B. Submission Flow (Data Tagging)
When a data collector opens a form at a center, the system automatically captures the origin:
1. **URL Parameter:** `https://haloform.com/fill/form_id?project=LON_2024&site=BPA&center=camp_A`
2. **Metadata Injection:** The UI reads the parameters and silently attaches them to the submission JSON.

C. Patient UI Contextualization (Dashboard Filtering)

To ensure patients only see relevant forms, the Patient Dashboard becomes "Context-Aware":
1. **Dynamic Filtering:** If a patient visits `patient.haloform.com?project=LON_2024&site=BPA`, the dashboard will only fetch and display forms belonging to that project and hospital.
2. **Context Locking:** The app saves the context in `localStorage`. If the patient refreshes the page or returns later, the app remains "locked" to that portal.
3. **Site-Specific Branding:** The UI can dynamically load the project's or hospital's logo, contact information, and welcome message based on the active context.

D. Database Storage Structure
Final submissions will be stored with a unified metadata block for better segregation and tracking:
```json
{
  "metadata": {
    "project_id": "proj_001",
    "hospital_id": "BPA",
    "center_id": "camp_A",
    "submitted_by": "user_123",
    "submitted_at": "2026-05-13T10:00:00Z"
  },
  "data": {
    "patient_age": 22,
    "bmi": 19.4
  }
}
```

E. QR Generation & Deployment Workflow
The Admin Panel serves as the central hub for generating physical access points for the field sites:
1. **Portal QRs:** A single QR code for a hospital under a project (e.g., `?project=LON_2024&site=BPA`) that directs patients to a filtered dashboard.
2. **Form-Specific QRs:** Deep-link QRs (e.g., `/fill/form_id?project=LON_2024&site=BPA`) that take the patient directly to a specific instrument while still preserving the site-tagging metadata.
3. **Automated Poster Generation:** The Admin UI will provide a "Print Site Poster" feature, generating a standardized PDF with the branding, instructions, and the encoded QR code for immediate physical deployment.

---

4. Key Benefits

Scalability & Management
Better scalability for multi-hospital deployments and easier management of nationwide/statewide programs. A new hospital can be simplified onboarded into an existing project.

Centralized Reporting & Data Segregation
Centralized project-level reporting and analytics. Better segregation and tracking of data by Project, Hospital, Center/Camp, and User.

Reusable Forms

Forms can be created once at the Project level and reused across all participating hospitals.

Data Isolation & Security
By tagging every row with `project_id` and `hospital_id`, we can easily implement **Row-Level Security**. A researcher from Ramaiah will only be able to query rows where `hospital_id == 'MSR'`.

---

5. Future Roadmap: PostgreSQL Integration
While SQLite works for local testing, the **PostgreSQL** migration will formalize these relationships using Foreign Keys:
* `hospitals.project_id` $\rightarrow$ `projects.id`
* `forms.project_id` $\rightarrow$ `projects.id`
* `submissions.hospital_id` $\rightarrow$ `hospitals.id`
* `submissions.project_id` $\rightarrow$ `projects.id`

This ensures that data cannot be orphaned and maintains the "Traceability Chain" required for clinical validation.

---

6. End-to-End Workflow: From Admin to Database

Step 1: Project Creation
BPA Admin creates a **Project** (e.g., a new nationwide program).
* **Input:** "Longevity Pilot 2026"
* **DB Result:** Row created in `projects` table.

Step 2: Hospital Mapping
Hospitals/Centers are mapped to the Project.
* **Input:** Add "Baptist Hospital Bangalore" (Code: `BPA`) to the Project.
* **DB Result:** Row created in `hospitals` table, linked to the `project_id`.

Step 3: Form Assignment
Forms are created and attached to the Project (or Hospital context). Hospitals use assigned forms within their project scope.
* **DB Result:** `forms` table row is created with `project_id`.

Step 4: QR Generation (Automatic)
Upon clicking **"Publish"** in the Form Builder:
1. The system constructs a **Context-Aware URL**:
   `https://patient.haloform.com/fill/5?project=LON_2026&site=BPA`
2. The Admin UI renders this as a downloadable **QR Code** and a printable **Site Poster**.

Step 5: Patient Submission
1. Patient scans the QR code.
2. The Patient App extracts `project=LON_2026` and `site=BPA` from the URL.
3. Upon submission, the app attaches these IDs, along with `center_id` and `submitted_by`, to the metadata.

4. **Backend Result:** A new row in `submissions` is created with all clinical data perfectly tagged for sorting by project, hospital, center, and user.

---

7. System Flow Diagram

